

Phong Pham

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EDUCATION

Gettysburg College, Gettysburg, PA

Dec 2025

B.Sc. Computer Science, Data Science Minor, Cum Laude (3.7/4.0)

PUBLICATIONS & PRESENTATIONS

Pham, P.T., Neller, T.W. (2026). FairKalah Without Empty Captures. In: *Advances in Computer Games* (CG 2026).

Maastricht University, Netherlands. [Accepted; in press] — *Nominated for Best Paper Award*

Neller, T.W., Nguyen, Q.H., Pham, P.T., Phan, L.T., Presser, C.G.M. (2024). Optimal Play of the Great Rolled Ones Game. In: Hartisch, M., Hsueh, CH., Schaeffer, J. (eds) *Advances in Computer Games*. ACG 2023. Lecture Notes in Computer Science, vol 14528. Springer, Cham. https://doi.org/10.1007/978-3-031-54968-7_5

RESEARCH EXPERIENCE

Research Assistant, Fulbright University – AI Institute

Aug 2026 – Present

PI: Dr. Quan Nguyen

- Joining a newly funded, 12-month AI Institute project that couples active search with online representation adaptation to improve rare-species discovery in ecological image collections.
- Support experiments, dataset expansion (including the Labeled Information Library of Alexandria: Biology and Conservation, LILA BC), and development of the final end-to-end system, across stages from codebase setup and baseline reproduction through representation-adaptation implementation, benchmarking, and manuscript preparation.

Undergraduate Researcher, Gettysburg College

Jun 2023 – Feb 2026

PI: Dr. Todd Neller

Project: Game-Theoretic Analysis and ML Modeling of Fair Starting Positions in Mancala

- Solved for fair starting configurations (positions where perfect play ends in a draw) in a classic combinatorial game by exhaustively searching a modified rule variant, building a 1.2GB endgame database to make exact game-tree search tractable at scale.
- Mined 63M+ unique optimal game states and 195M+ possible moves from perfect-play search trees to characterize strategic patterns (e.g., move branching factor, capture frequency, positional tension) across the game.
- Engineered 25+ domain-specific features and trained ML models (CatBoost, Random Forest, logistic regression) to predict move quality and game outcome from board state, reaching up to 94% accuracy predicting optimal moves and cutting prediction error by ~89% versus a linear baseline, generalizing across 10M+ held-out states.
- Wrote first-author paper and presented results at Computers and Games conference in the Netherlands (2026).

Project: Solving Optimal Strategy for a Stochastic Dice Game via Dynamic Programming

- Formulated a two-player dice game (Great Rolled Ones) as a Markov decision process and derived closed-form recursive win-probability equations over a bounded, high-dimensional state space (player scores, turn total, risk state).
- Solved for the exact optimal policy using value iteration to convergence, then used the resulting probabilities to compute a game-theoretically fair compensation ("komi") adjustment, closing a ~10% second-player advantage to within 1%.
- Designed several human-playable policies, from single-rule baselines to case-based strategy. Quantified each one's win rate gap over optimal play (5.4% down to 1%), balancing decision complexity against performance.
- Co-authored paper and presented results at the 2023 Advances in Computer Games [conference](#).

Senior Capstone Project: Impact Rehab MSK Clinical Assessment System

2025

- Designed and implemented a clinical musculoskeletal assessment platform that digitized patient evaluations into a secure, structured workflow supporting functional movement assessments.
- Engineered an end-to-end clinical data pipeline for patient intake, automated score computation, longitudinal data storage, and standardized reporting to support clinical interpretation.
- Designs were implemented into production within the clinic and reduced manual workloads by 20 hrs per week.

Undergraduate Researcher, Gettysburg College

2023

PI: Dr. Sunghee Kim

Self-directed study: Sentiment Classification for Vietnamese Text

- Independently learned core ML/AI methods (regression, classification, neural networks) and optimization fundamentals with no prior coursework background.
- Applied this to implement an LSTM-CNN neural network in Python for 3-class sentiment classification of Vietnamese text, replicating an existing modeling approach as a foundation for later research.

PROFESSIONAL EXPERIENCE

Intern, DIGINET Corporation

2022

- Built a JavaScript library converting MSSQL database schemas to MySQL from the ground up, including type mapping across 20+ MSSQL data types (e.g., varbinary, uniqueidentifier, datetimeoffset) with length-dependent branching logic (e.g., varchar → TEXT vs. LONGTEXT by size threshold).
- Handled edge cases in reserved-keyword collisions, invalid SQL identifier characters, and semantic translation of MSSQL system defaults (e.g., getdate(), newid()) to MySQL equivalents.

TEACHING EXPERIENCE

Teaching Assistant, Gettysburg College

Sep 2022 – Dec 2025

- Courses: Intro to Computer Science, OOP and Calculus I, II, III.

AWARDS & HONORS

Best Paper Award Nominee, Computers and Games (CG) Conference

2026

Dean's Honor List, Gettysburg College

2022, 2023, 2024, 2025

CS Departmental Honors, Gettysburg College (top 3 students selected)

2025

Presidential Scholarship, Gettysburg College

2022-2025

LEADERSHIP & SERVICE

President, Vietnamese Student Association (VSA), Gettysburg College

2023-2024

First Class, Boy Scouts of Vietnam

2009-2021

TECHNICAL SKILLS

Programming: Python, Java, MATLAB, JavaScript, HTML, SQL, Dart, PHP, C++, RESTful API, OpenGL

Machine Learning: PyTorch, Scikit-learn, Pandas, NumPy, Feature Engineering, Model Evaluation, Neural Nets, Classification